

Task No.: 1**Date: 31-12-2025**

Title: Develop an application that uses GUI components, Font and Colors**Aim:**

To Develop an Android application that displays a text message “**Hello World**” on the screen.

The application should contain **two buttons** to perform the following operations:

1. Change Font Size Button

- When the button is clicked, the font size of the text “**Hello World**” should increase.

2. Change Font Color Button

- When the button is clicked, the font color of the text “**Hello World**” should change.

Code:*MainActivity.kt*

```
package com.example.madtask2
```

```
import android.graphics.Color
import android.os.Bundle
import android.widget.Button
import android.widget.TextView
import androidx.appcompat.app.AppCompatActivity
```

```
class MainActivity : AppCompatActivity() {
```

```
    private val fontSizes = arrayOf(20f, 24f, 28f, 32f, 36f)
    private var fontIndex = 0
```

```
    private val fontColors = arrayOf(
        Color.RED,
        Color.BLUE,
        Color.GREEN,
```

```

        Color.MAGENTA,
        Color.BLACK
    )
    private var colorIndex = 0

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        val helloText = findViewById<TextView>(R.id.helloText)
        val btnFontSize = findViewById<Button>(R.id.btnFontSize)
        val btnFontColor = findViewById<Button>(R.id.btnFontColor)

        btnFontSize.setOnClickListener {
            helloText.textSize = fontSizes[fontIndex]
            fontIndex = (fontIndex + 1) % fontSizes.size
        }

        btnFontColor.setOnClickListener {
            helloText.setTextColor(fontColors[colorIndex])
            colorIndex = (colorIndex + 1) % fontColors.size
        }
    }
}

```

activity_main.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:background="@android:color/white">

    <TextView
        android:id="@+id/studentInfo"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Name: Yoga Shri P.K\nRoll No: 23BCS192"
        android:textSize="18sp"
    >

```

```
android:textStyle="bold"  
android:textColor="@android:color/black"  
android:gravity="center"  
android:padding="16dp" />
```

```
<LinearLayout
```

```
    android:layout_width="match_parent"  
    android:layout_height="0dp"  
    android:layout_weight="1"  
    android:orientation="vertical"  
    android:gravity="center">
```

```
<TextView
```

```
    android:id="@+id/helloText"  
    android:layout_width="wrap_content"  
    android:layout_height="wrap_content"  
    android:text="Hello World"  
    android:textSize="24sp"  
    android:textColor="@android:color/black"  
    android:layout_marginBottom="24dp" />
```

```
<Button
```

```
    android:id="@+id/btnFontSize"  
    android:layout_width="wrap_content"  
    android:layout_height="wrap_content"  
    android:text="Change Font Size" />
```

```
<Button
```

```
    android:id="@+id/btnFontColor"  
    android:layout_width="wrap_content"  
    android:layout_height="wrap_content"  
    android:text="Change Font Color"  
    android:layout_marginTop="12dp" />
```

```
</LinearLayout>
```

```
</LinearLayout>
```

Output:

Name: Yoga Shri P.K
Roll No: 23BCS192

Name: Yoga Shri P.K
Roll No: 23BCS192

Name: Yoga Shri P.K
Roll No: 23BCS192

Hello World

Change Font Size

Change Font Color

Hello World

Change Font Size

Change Font Color

Hello World

Change Font Size

Change Font Color

Name: Yoga Shri P.K
Roll No: 23BCS192

Name: Yoga Shri P.K
Roll No: 23BCS192

Hello World

Change Font Size

Change Font Color

Hello World

Change Font Size

Change Font Color

Result:

An android application that displays a “Hello World!” with two buttons are created with functionality of changing the font size and font color is executed successfully.